Colouring minimal Cayley graphs: the road from Babai's conjectures to their resolution

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Abstract

In 1978 Babai conjectured that minimal Cayley graphs have a bounded chromatic number. In 1995 he changed his mind and conjectured the opposite.

1 AI/Data usage

Used Claude to generate some TikZ's code with prompt: *Give me standalone TikZ code for a Venn-style containment diagram: one large circle labeled 'no-lonely-color graphs' fully containing a smaller circle labeled 'minimal Cayley graphs', to show the proper-subset relationship. Use plain black outlines, light gray fill on the inner circle, and make sure the outer label doesn't overlap the inner circle. Compile it to check it renders before sending it to me.*

2 Introduction

3 Preliminaries

3.1 Standard notation for paper

3.2 Cayley graphs with small examples

3.3 Minimal generating sets

3.4 Hypergraph coloring

4 Babai's question, 1978-1995

4.1 Minimal Cayley graphs are no-lonely-color

4.2 Forbidden subgraphs

4.3 Babai's 1978 conjectures

Both conjecture 2.2 and 3.5 from Babai's 1978 paper

4.4 The 1995 counter-conjecture

5 Evidence in both directions

5.1 Abelian groups are 3-colorable

5.2 More positive results

5.3 Negative results

6 Necessary tools

7 No-lonely-color graphs of large chromatic number

7.1 Tranquil hypergraphs

7.2 Tranquility of C_d

7.3 The construction

7.4 The proof

8 Minimal Cayley graphs of large chromatic number

9 Open questions

10 Appendix

References